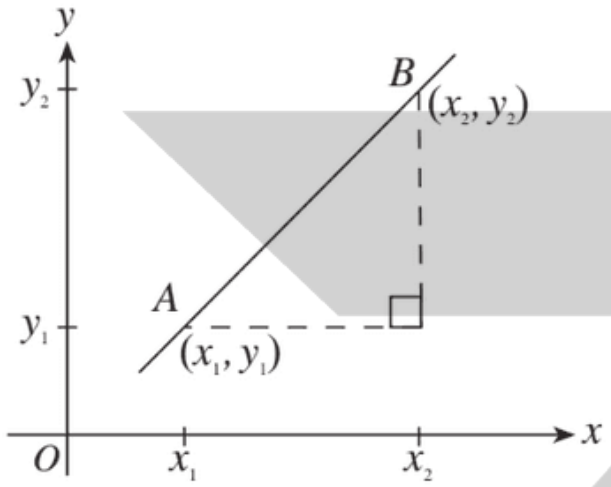
1. Coordinate Geometry

1.1. Distance formula

- The distance formula helps us find the distance, or length, between two points.

Let us consider two points, on the same linear line, $A(x_1, y_1)$ and $B(x_2, y_2)$. The lines $x = x_2$ and $y = y_1$ can be drawn, resulting in a **right angled triangle** with the line AB as the **hypotenuse**.



The length of the **base** is obtained from $x_2 - x_1$ and the **height** is obtained from $y_2 - y_1$.

- Using Pythagoras theorem, we can derive the **distance formula** for two points, A and B, as:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Mid-point formula

- The mid-point formula helps us find a point, usually denoted by M, which is **equidistant** from two other points.The mid-point of two points can be found using:

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Example 1

The distance between points $A(0, 4)$ and $B(a, 1)$ is 5 units. Find the value of the **positive integer** a.

Answer

Using the distance formula:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

We can substitute our values in

$$5^2 = (a - 0)^2 + (1 - 4)^2$$

$$a^2 = 16 \implies a = 4$$

Example 2

A line segment, joining points $A(2, -3)$ and $B(4, a)$ has a midpoint $M(b, -2)$. Find the values of a and b.

Answer

Using the mid-point formula

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

The y value of the mid-point can be represent as

$$y_M = \frac{y_1 + y_2}{2}$$

$$-2 = \frac{-3 + a}{2}$$

$$a = -1$$

The x value of the mid-point can be represented as

$$x_M = \frac{x_1 + x_2}{2}$$

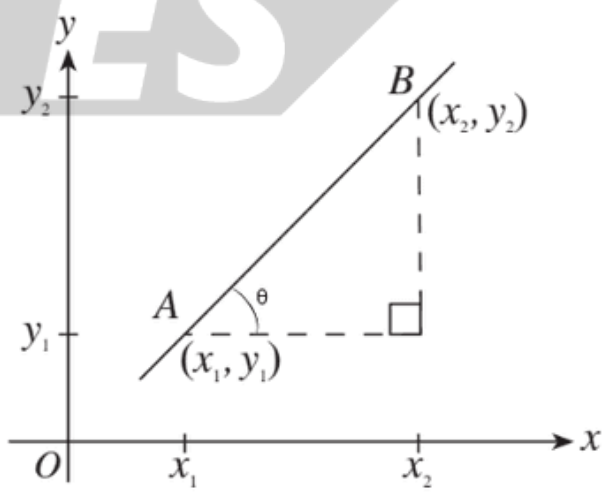
$$b = \frac{2 + 4}{2}$$

$$b = 3$$

1.2. The Gradient

- The gradient, also called the slope, describes the **inclination** of a line. It gives a numerical value to the **angle** a line makes with the **positive** x-axis.
  - It is popularly denoted by m in the equation $y = mx + c$.

Consider two points, $A(x_1, y_1)$ and $B(x_2, y_2)$. A right angled triangle can be drawn, with a similar explanation to the one in the distance formula, with the hypotenuse line AB .



We can apply the trigonometric ratio

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$$tan(\theta) = \frac{\text{length of opposite side}}{\text{length of adjacent side}}$$

In the context of the diagram above

$$tan(\theta) = \frac{y_2 - y_1}{x_2 - x_1}$$

We will also denote m as

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

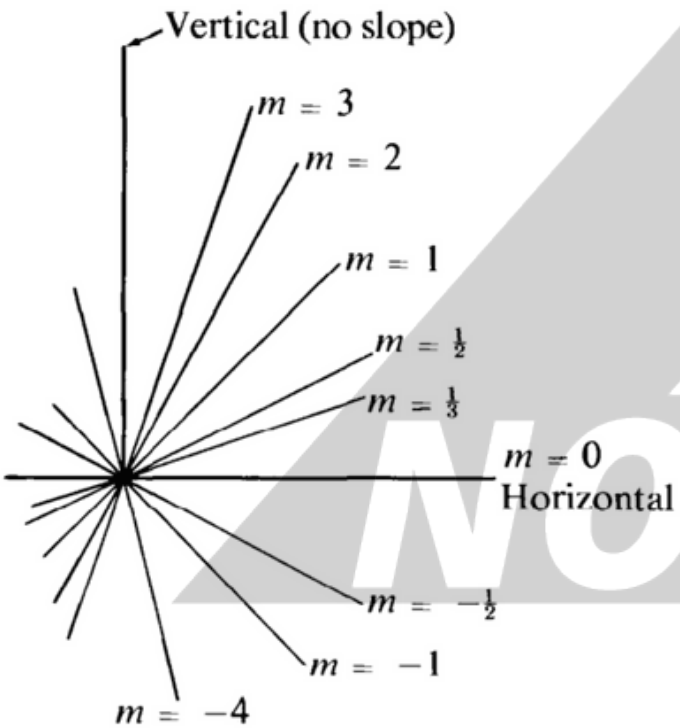
Such that we are able to find the angle of inclination, θ, by:

$$tan(\theta) = m$$

$$\theta = tan^{-1}(m)$$

Magnitude of the Gradient

- The function is going to have an **increasing** y value, as the x value increases, if the gradient has a **positive** value. This means the curve is moving upwards.
- The function is going to have a **decreasing** y value, as the x value increases, if the gradient has a **negative** value. This means the curve is moving downwards.



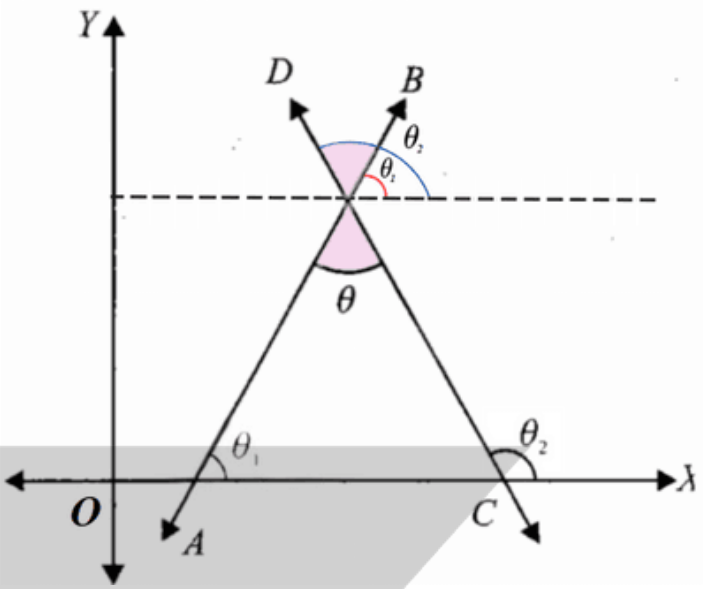
Angle between two lines

The angle, θ, between two lines having gradients m_1 and m_2 is given by

$$tan(\theta) = \pm\frac{m_2 - m_1}{1 + m_1 m_2} \implies \theta = tan^{-1}\left(\pm\frac{m_2 - m_1}{1 + m_1 m_2}\right)$$

- Note that you must take the positive value of the ratio if you are trying to find the **acute angle** between two lines.

Another way to look at this formula is by considering the formula to find θ using the gradient m



Let the line AB have a gradient of m_1 and the line DC have a gradient of m_2 such that

$$tan(\theta_1) = m_1 \implies \theta_1 = tan^{-1}(m_1)$$

$$tan(\theta_2) = m_2 \implies \theta_2 = tan^{-1}(m_2)$$

From the diagram above

$$\theta_2 - \theta_1 = \theta$$

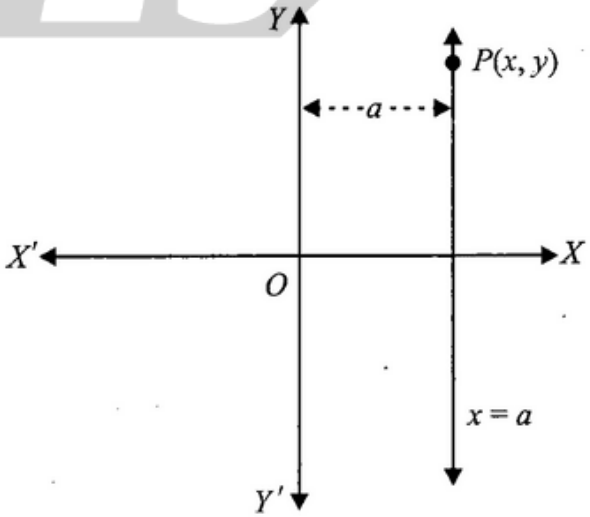
This gives us

$$tan^{-1}(m_2) - tan^{-1}(m_1) = \theta$$

1.3. Equations of lines

Equation of line parallel to y-axis

- The equation of a line parallel to the y-axis can be written in the form $x = a$.
  - All points on this line have the coordinate $P(a, y)$.



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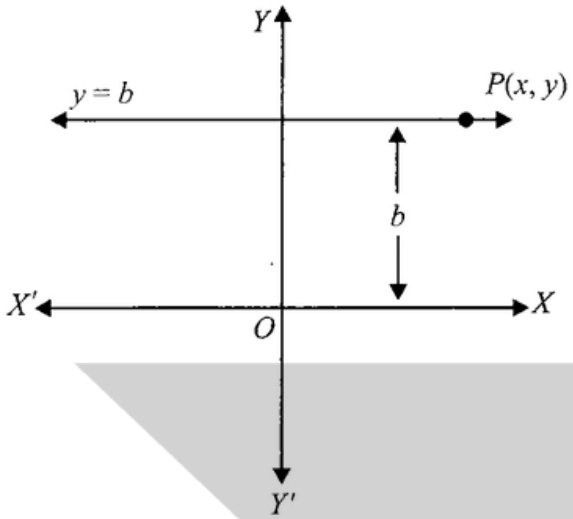
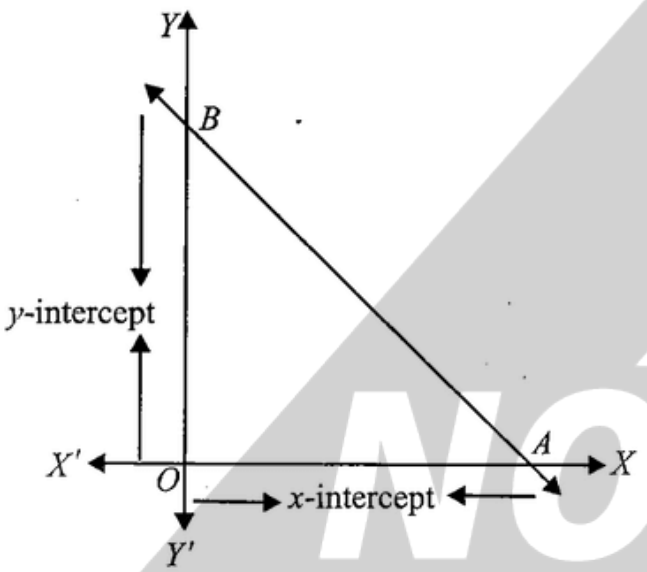
Equation of line parallel to x-axis

- The equation of a line parallel to the x-axis can be written in the form $y = b$.
  - All points on this line have the coordinate $P(x, b)$.



Intercepts

- An intercept is the point at which a line **cuts** the coordinate axes.



For this graph:

- The y-intercept is $(0, B)$
- The x-intercept is $(A, 0)$

Equation of a straight line

The equation of a straight line is given by

$$y = mx + c$$

- m represents the gradient.
- c represents the y-intercept $(0, c)$. It can also be looked at in terms of a **vertical translation**.

Point-slope formula

The point-slope formula is **another way** to find the equation of a straight line.

- Consider a general point $P(x, y)$, which can be **any point** on the line, and a **known point** $Q(x_1, y_1)$ which the line passes through.

The gradient, m, can be found using the formula

$$m = \frac{y - y_1}{x - x_1}$$

From this, we find the point-slope formula by making (y - y_1) the subject of the formula

$$(y - y_1) = m(x - x_1)$$

Example

Find the equation of the straight line that passes through the points (-5, 3) and (-4, 1).

Answer

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 3}{-4 - (-5)}$$

$$m = -2$$

$$y = -2x + c$$

Now we can substitute any point in this equation to find the value of the y-intercept c

$$1 = -2(-4) + c$$

$$c = -7$$

So our final answer is

$$y = -2x - 7$$

1.4. Relation between lines

Relation between parallel lines

- Parallel lines have the **same gradients**. This is due to the angle between them being 0° degrees.

Consider two lines, $y = m_1x + c$ and $y = m_2x + c$. Given that these two lines are **parallel**, we can say:

$$m_1 = m_2$$

Relation between perpendicular lines

- Perpendicular lines create a 90° degree angle between them upon intersection. They are also called normal lines.

Consider two lines, $y = m_1x + c$ and $y = m_2x + c$. Given that these two lines are **perpendicular**, we can say:

$$m_1 \times m_2 = -1$$

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Which means it is possible to find the gradient of one of the lines, given the gradient of the other line

$$m_2 = -\frac{1}{m_1} \text{ or } m_1 = -\frac{1}{m_2}$$

1.5. Perpendicular bisector

- A perpendicular bisector is a line that bisects another line **perpendicularly**, at a 90 degree angle, at its **midpoint**.

Example

Two points have coordinates $A(5,7)$ and $B(9,-1)$. Find the equation of the perpendicular bisector of the line AB.

Solution

The gradient, m_1, of AB can be calculated using

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-1-7}{9-5}$$

$$m = -2$$

We can also find the gradient of the perpendicular, m_2, using

$$m_1 \times m_2 = -1$$

$$-2 \times m_2 = -1 \implies m_2 = \frac{1}{2}$$

So we have a line equation, for the perpendicular bisector, in the form

$$y = \frac{x}{2} + c$$

To find the value of the y-intercept, c, we can use the fact that the perpendicular bisector **must** intersect the **midpoint** of the line AB.

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$M = \left(\frac{5+9}{2}, \frac{7-1}{2}\right)$$

$$M = (7,3)$$

Using these coordinates into our line equation, we get

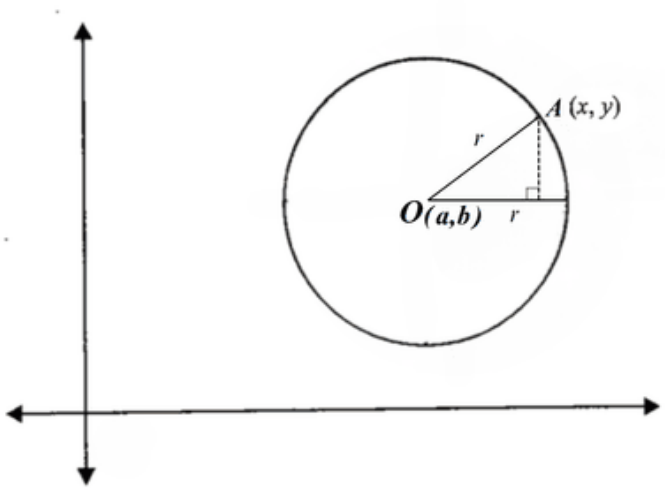
$$3 = \frac{7}{2} + c \implies c = -\frac{1}{2}$$

So the equation of the perpendicular bisector is

$$y = \frac{x}{2} - \frac{1}{2}$$

1.6. Equation of a circle

- The equation of a circle helps us describe a circle on the coordinate plane.



- Consider a circle with centre $O(a,b)$ and radius OA, equal to r.
  - A perpendicular line from $A(x,y)$ can be drawn to depict a right angled triangle.
- This right angled triangle has:
  - r as its hypotenuse
  - $(x-a)$ as its base
  - $(y-b)$ as its height

Using the Pythagoras theorem

$$r^2 = (x-a)^2 + (y-b)^2$$

- Where the coordinates of the centre, O, of this circle are (a,b)

The equation of a circle can be represented in different ways, and it is important to correctly identify the coordinates of the centre from them.

Equation of circle	Coordinates of centre O
$r^2 = (x-a)^2 + (y-b)^2$	(a,b)
$r^2 = (x+a)^2 + (y+b)^2$	$(-a,-b)$
$r^2 = (x+a)^2 + (y-b)^2$	$(-a,b)$
$r^2 = (x-a)^2 + (y+b)^2$	$(a,-b)$

- A point, (x,y), lies on the circumference of the circle if it satisfies the equation of the circle when inputted.
  - Which means inputting points x and y into the equation of the circle will give you the value of r^2 if it lies on the **circumference**.

General equation of a circle

The expanded form of the equation of the circle is also known as the general equation of the circle.

$$0 = x^2 - 2ax + y^2 - 2by + c$$

Where $c = a^2 + b^2 - r^2$

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Centre	(a, b)
Radius	$r = \sqrt{a^2 + b^2 - c}$

If the general form is given as

$$0 = x^2 + 2ax + y^2 + 2by + c$$

Centre	$(-a, -b)$
Radius	$r = \sqrt{a^2 + b^2 - c}$

- Another way to find the centre and radius is by **completing the square**, shown in the example 2 below.

Example 1

Points $A(-3, 4)$ and $B(3, -4)$ lie on the circle, and the line AB is the diameter of that circle. Find the equation of the circle.

Answer

Since the diameter is the line AB, its midpoint must be the coordinates of the centre O of the circle.

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$O = \left(\frac{-3 + 3}{2}, \frac{4 - 4}{2}\right)$$

$$O = (0, 0) \text{ [This means the centre lies on the origin]}$$

As the points A and B lie on the circle, they must also satisfy the equation of the circle when inputted. Using this fact, we can input one of their coordinates to find the radius. Here we use the coordinates of A.

$$r^2 = (x - a)^2 + (y - b)^2$$

$$r^2 = (-3 - 0)^2 + (4 - 0)^2$$

$$r^2 = 25 \implies r = 5$$

Thus, the equation of the circle can be written as

$$x^2 + y^2 = 5^2$$

Example 2

Find the center and the radius of the circle with the equation $x^2 + y^2 - 10x - 8y - 40 = 0$.

Answer

Given the equation of the circle, we can complete the square for x and y to get

$$(x - 5)^2 - 25 + (y - 4)^2 - 16 - 40 = 0$$

$$(x - 5)^2 + (y - 4)^2 = 81$$

$$(x - 5)^2 + (y - 4)^2 = 9^2$$

The centre of the circle is $(5, 4)$ and the radius is 9.

1.7. Translation applied to a Circle Equation

- A translation applied to a circle equation moves the coordinates of the centre and all the points on the circumference.
  - It does not change the radius, r.

Consider a circle equation

$$r^2 = (x - a)^2 + (y - b)^2$$

With coordinates of the centre as (a, b). To apply a translation of $\binom{c}{d}$, we can add this to the coordinates of the centre.

$$(a + c, b + d)$$

So the equation of the transformed graph is

$$r^2 = (x - [a + c])^2 + (y - [b + d])^2$$

1.8. Intersection of lines and circles

- when an equation in the form of $ax^2 + bx + c = 0$ is created from the equation of the line and the circle, the value of $b^2 - 4ac$ decides the number of intersection points
- if the value of $b^2 - 4ac$ is
  - $>$ 0 then two distinct points of intersection are formed
  - = 0 then a single point of intersection (line is a tangent to the circle) is formed
  - < 0 then there are no points of intersection

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